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Programmable RNA translation through deep learning-driven IRES discovery and de novo generation

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The precise control of protein expression is a major bottleneck in the development of RNA therapeutics. Internal ribosome entry sites (IRES) overcome traditional limitations by enabling cap-independent translation initiation, making them highly desirable tools for synthetic biology and therapeutic payload expression. However, the complex structure-function relationship of IRES elements has historically hindered their rational design. Here we show that a comprehensive, end-to-end artificial intelligence framework unifies IRES identification, evolutionary optimization and de novo generation. First, IRES-LM, an ensemble of two language models trained on 46,774 sequences, predicts linear mRNA IRESs with a 15% improvement in area under the curve and F1 score over existing methods. In addition, IRES-LM demonstrates remarkable cross-applicability to circular RNA IRESs, correctly identifying all 21 experimentally validated circular RNA IRESs and outperforming benchmark methods. Next, IRES-EA integrates an evolutionary algorithm with IRES-LM to induce IRES functionality through targeted mutations. Computational evaluation of 37,293 non-IRES sequences showed 60% predicted functional conversion, with large-scale massively parallel reporter assay validation of 12,000 mutated sequences demonstrating 98.4% acquired IRES functionality, confirming both computational predictions and experimental functionality. Further, IRES-DM employs a diffusion model to de novo generate novel IRES sequences that outperform the state-of-the-art method. Massively parallel reporter assay validation using another set of 12,000 IRES-DM-generated sequences revealed 99.3% detectable IRES functionality. Notably, IRES-DM shows diverse generation capacity, producing sequences ranging from natural-like candidates to structurally conserved yet sequence-divergent designs. Motif analysis revealed both natural-prevalent and design-enriched high-activity motifs. Together, this framework establishes a robust approach for programmable RNA translation, expanding the molecular toolkit for scaling up next-generation biomedical discovery and RNA-based therapeutics.

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Internal ribosome entry sites (IRESs) are unique *cis*-acting RNA elements found in both mRNA and circular RNA (circRNA) that enable ribosome recruitment and translation initiation without requiring a 5′ cap structure. This cap-independent translation mechanism is particularly crucial during conditions where conventional cap-dependent translation is inhibited, such as cellular stress or viral infection. IRESs represent important therapeutic targets for regulating protein expression in viral infections and cellular processes, with potential for manipulation using small-molecule inhibitors[1–3]. Despite their importance, the precise roles of RNA structures, IRES *trans*-acting factors and RNA-binding proteins in IRES function remain incompletely understood[4–6], contributing to a fundamental challenge in IRES research: determining whether all IRESs share common mechanisms, sequences or structural features[7]. This uncertainty stems from the limited number of experimentally confirmed IRESs, making their computational identification challenging. Furthermore, the ability to design and optimize IRES elements is crucial for therapeutic applications and synthetic biology, particularly for enhancing RNA vaccine efficacy[8,9]. Engineered IRESs can potentially enable targeted control of gene expression in the absence of 5′ cap, addressing a critical need in biomedical research. These challenges necessitate innovative computational approaches for accurate identification, characterization, directed mutation and de novo design of IRES elements, which is the central focus of our study.

The traditional methods of IRES identification are time-consuming and labour-intensive, resulting in a restricted dataset of confirmed IRESs. This limitation has hindered comprehensive studies of their common features and functions, creating a major barrier to advancing our understanding of translation regulation. Current bioinformatics tools such as IRESPred[10], IRESfinder[11] and IRESpy[12] employ various algorithms such as SVM, logit model and XGBoost to predict IRESs in linear mRNA, whereas DeepCIP[13] uses deep neural networks specifically for circRNA IRES prediction. However, these approaches rely heavily on manually crafted features such as *k*-mers and fail to capture the universal characteristics across diverse IRES types. Moreover, they demonstrate limited cross-applicability between linear mRNA and circRNA contexts[13], further restricting their utility in comprehensive IRES research. To address these limitations, we developed IRES-LM by systematically evaluating multiple RNA foundation models and selecting UTR-LM[14] and RNA-FM[15] based on their superior performance on translation-related tasks (Figs. 1a and 2a). UTR-LM is the state-of-the-art language model focused on 5′ untranslated regions (UTR), whereas RNA-FM excels in non-coding RNA analysis and has demonstrated outstanding performance with 5′ UTRs. Training on 46,774 binary-labelled sequences (9,172 linear mRNA IRESs and 37,602 non-IRESs) from Weingarten-Gabbay et al.[16], IRESbase[17], IRESite[18], Rfam[19] and IRES-LM learns IRES-specific patterns directly from sequence data and captures biologically relevant features without explicit supervision. In rigorous benchmarking against IRESfinder[11], IRESpy[12] and DeepCIP[13], IRES-LM demonstrates superior performance, achieving a 15% improvement in area under the curve (AUC) and F1 score over the state-of-the-art performance. Moreover, IRES-LM exhibits remarkable cross-applicability by correctly identifying all 21 experimentally validated circRNA IRESs, outperforming existing methods[11–13].

Engineering IRES functionality is essential for enhancing translation efficiency with polycistronic design, particularly for improving the effectiveness of mRNA and circRNA vaccines[8,9]. Previous research has shown that rational manipulation of IRESs can increase protein yields several hundred-fold[8]. Building on our IRES-LM identification model, we develop IRES-EA, an evolutionary algorithm framework designed for mutation-guided induction or disruption of IRES functionality while preserving secondary structure (Figs. 1b and 2b). IRES-EA treats original sequences as 'wild-type' and systematically optimizes them into 'mutated-type' variants with modified IRES functionality. The algorithm offers three strategies for selecting mutation sites: random nucleotides, critical nucleotides for IRES functionality and custom-defined nucleotides. In extensive computational evaluation with 37,293 non-IRES sequences, IRES-EA successfully converted 60% into IRES sequences as predicted by IRES-LM, with 5% achieving high IRES probability exceeding 90%, allowing up to nine mutation sites per sequence. Bicistronic reporter assays on EMCV and CVB3 IRES variants showed that all tested mutants retained IRES activity, with secondary structure analysis revealing that high-activity variants maintain wild-type-like RNA folding patterns. Large-scale massively parallel reporter assay (MPRA) (Fig. 1) validation of 12,000 mutated sequences further demonstrated that 98.4% acquired detectable IRES functionality, directly confirming IRES-EA's conversion capability and IRES-LM's prediction accuracy.

Although IRES-EA focuses on mutation-guided functional induction from existing sequences, de novo design of IRES sequences represents a crucial approach for expanding the library of translation regulatory elements. To address this need, we developed IRES-DM, which implements a denoising diffusion probability model (DDPM)[20] to generate novel IRES sequences (Figs. 1c and 2c). IRES-DM generates entirely new IRES sequences from random noise, without requiring predefined templates. We implemented two versions (174 nt fixed-length and ≤200-nt variable-length) with two training strategies each: a reward-guided approach and a direct-training approach. To benchmark IRES-DM's generative capability, we compared variable-length IRES-DM against GenerRNA[21], a state-of-the-art RNA generative model, demonstrating superior performance in generating high-quality, non-duplicated sequences across multiple filtering criteria. Fixed-length IRES-DM outperformed random sequence generation across all quality metrics, confirming its ability to learn IRES-specific patterns rather than generating random outputs. Large-scale MPRA validation of 12,000 generated sequences revealed that 99.3% exhibited functional IRES activity, with consistent performance across both model versions and training strategies, confirming IRES-DM's robust de novo generation capability. We also tested two generated sequences similar to VCIP IRES, both of which demonstrated obvious IRES activity compared with the negative control. Most strikingly, IRES-DM can generate sequences sharing only 27.6% sequence similarity with natural IRESs such as BiP IRES[22] while maintaining remarkably similar secondary structures, revealing multiple viable evolutionary paths to achieving IRES functionality. In addition, analysis of experimentally validated sequences identified high-activity *k*-mer motifs, including patterns rare in natural IRESs, suggesting exploration of sequence space underutilized by natural evolution.

Our comprehensive framework, consisting of IRES-LM, IRES-EA and IRES-DM, enables precise identification, mutation effect identification and de novo generation of IRES sequences. Experimental validation confirms the functional accuracy of our computational predictions, with multiple engineered IRES variants demonstrating activity comparable to or exceeding wild-type sequences. These advances highlight the potential of our integrated approach to enhance IRES research and applications, with important implications for improving RNA vaccine efficacy and developing novel therapeutic strategies through precise control of protein expression.

Results

IRES-LM identifies IRES accurately

Recognizing IRES is crucial for understanding and manipulating translation regulation in both basic research and therapeutic applications. However, existing computational methods, such as IRESfinder[11], IRESpy[12] and DeepCIP[13], rely heavily on manual feature engineering and show limited performance in identifying diverse IRES elements. To address these challenges, we developed IRES-LM, a language model-based approach that leverages transfer learning from RNA foundation models to learn IRES-specific features directly from sequence data (Fig. 2a).

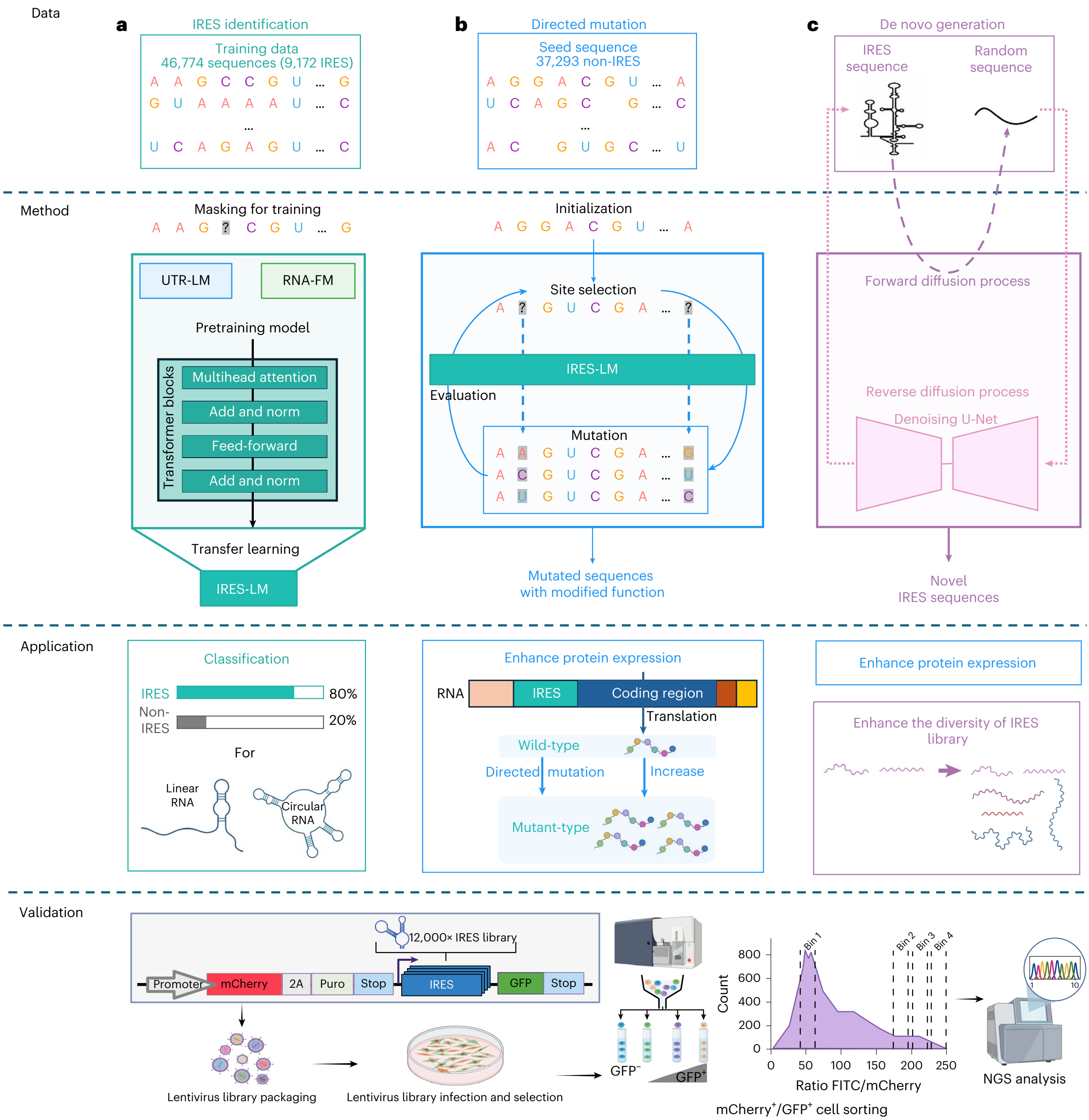

Fig. 1 | Comprehensive IRES framework for identification, directed mutation, and de novo generation of IRES elements. a, IRES identification. IRES-LM employs transfer learning from UTR-LM and RNA-FM pretrained models to identify IRES elements from 46,774 training sequences through a transformer architecture. **b**, Directed mutation. IRES-EA converts non-IRES sequences to functional IRES through iterative site selection, mutation and evaluation guided by IRES-LM, enabling enhanced protein expression. **c**, De novo generation. IRES-DM generates novel IRES sequences using a diffusion model with forward noise addition and reverse denoising processes, expanding the diversity of the IRES library while maintaining functional characteristics. Applications of this framework include IRES classification for both linear and circRNA and enhancing protein expression for RNA-based therapeutics. The IRES-EA and IRES-DM are validated by the large-scale MPRA workflow, which employs a dual-fluorescence bicistronic reporter measuring IRES-dependent GFP expression relative to constitutive mCherry. Following lentiviral packaging and transduction, FACS separated cells into four bins based on GFP/mCherry ratios: bin 1 (negative, GFP−) and bins 2–4 (positive with progressively increasing activity, GFP+). Next-generation sequencing (NGS) was then used to quantify sequence abundance across bins. Figure created in BioRender; Chu, Y. https://biorender.com/5y6q7k5 (2026).

We selected UTR-LM[14] and RNA-FM[15] as foundation models for IRES identification, as both have demonstrated excellence in 5′ UTR-related tasks where most IRESs are located. UTR-LM was specifically designed for 5′ UTR sequences; RNA-FM, although pretrained on diverse non-coding RNAs, demonstrated successful transfer to 5′ UTR function prediction tasks. To validate this selection, we compared their performance with two other prominent RNA foundation models: RNA-BERT, which focuses on structural alignment, and

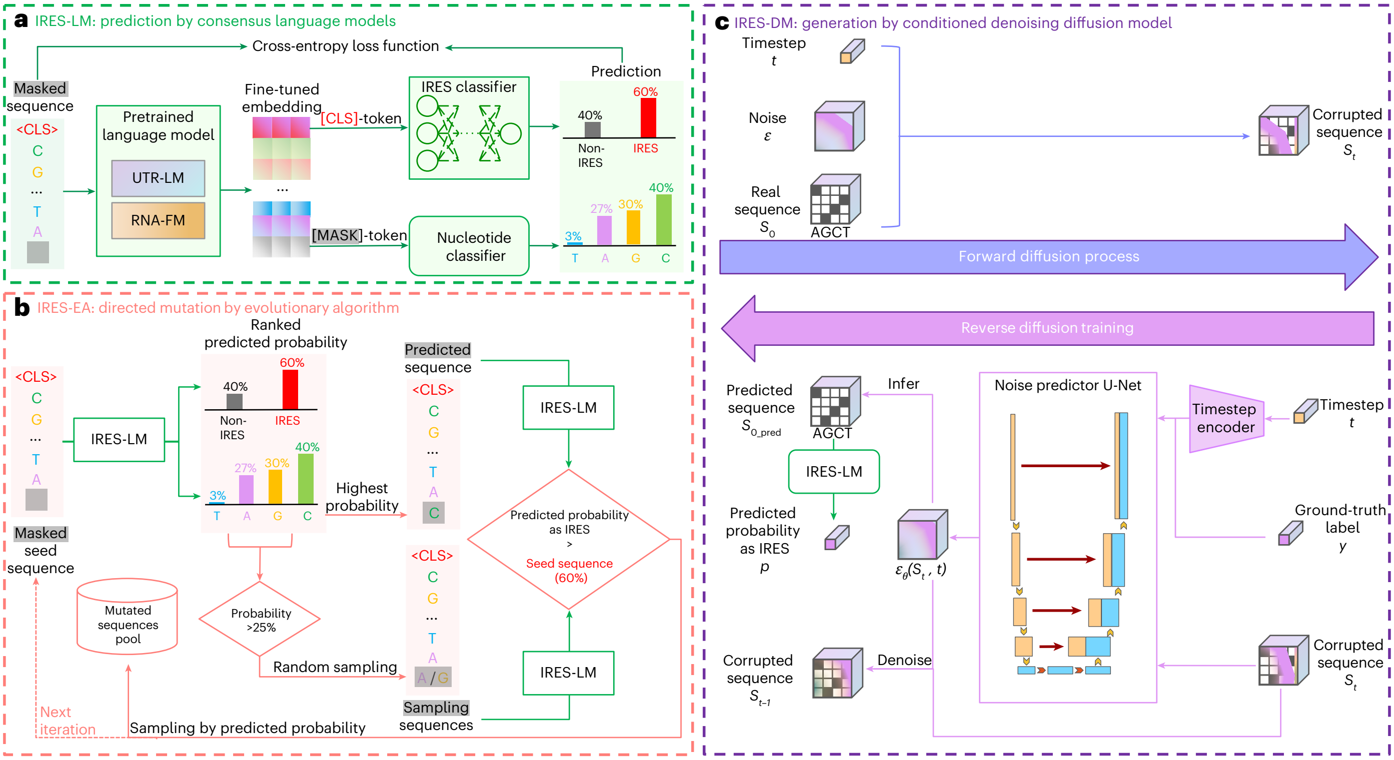

Fig. 2 | Overview of IRES prediction and generation methods. a, IRES-LM for IRES identification. A masked RNA sequence is input into a pretrained language model, comprising UTR-LM and RNA-FM. The output shows the probabilities of IRES and non-IRES to identify IRES elements within the sequence. **b**, IRES-EA for directed mutation. The iterative process of sequence mutation and optimization to achieve the desired IRES. The process consists of masking the seed sequence, generating mutated sequences according to the predicted probability from IRES-LM and using the sampled sequences as seed sequences for the next iteration. **c**, IRES-DM for novel IRES generation. The forward diffusion process starts with a real sequence converted into a one-hot encoded vector and then progressively adds Gaussian noise until the vector is completely noisy. The reverse denoising process involves training a neural network to gradually remove the noise, transforming pure noise back into a real sequence.

ERNIE-RNA, which incorporates base-pairing constraints. We evaluated all four models across four training strategies: random initialization, frozen pretrained model with trainable predictor, last-layer fine-tuning and full fine-tuning (Extended Data Fig. 1). UTR-LM showed remarkable adaptability, with area under the precision-recall curve (AUPR) improving from 0.31 (frozen) to 0.58 (full fine-tuning), an 87% relative improvement. RNA-FM maintained consistently high performance across all strategies, achieving the best individual model AUPR of 0.62. RNA-BERT demonstrated limited transfer learning benefits despite showing improvement from frozen to fine-tuned states, reaching only 0.53 AUPR. Uniquely, ERNIE-RNA exhibited negative transfer, with AUPR deteriorating from 0.35 (frozen) to 0.22 (fine-tuned), suggesting severe incompatibility between its base-pairing-focused pretraining and IRES identification requirements. The superior performance of fine-tuning over random initialization for UTR-LM and RNA-FM confirms that translation-relevant pretraining provides crucial advantages for IRES identification. Summary of optimal performance across all models (Fig. 3a) further validated our selection, with RNA-FM (AUPR of 0.62) and UTR-LM (AUPR of 0.58) substantially outperforming RNA-BERT (AUPR of 0.53) and ERNIE-RNA (AUPR of 0.22), demonstrating that alignment between pretraining objectives and target tasks determines transfer learning effectiveness. To further validate the importance of task-specific pretraining, we also evaluated Evo-2-1b-base, a general genomic foundation model with 25 blocks trained on over 9 trillion nucleotides of diverse genomic sequences. Given its massive scale, we adopted two evaluation strategies: extracting embeddings from different blocks with frozen parameters, and fine-tuning the last N blocks along with the downstream head. Despite its powerful general modelling capabilities, Evo-2 performed poorly on IRES identification (AUPR of 0.31), though marginally better than the structurally-focused ERNIE-RNA (AUPR of 0.22), suggesting that general sequence modelling provides some basic transferability compared with irrelevant specialized pretraining but still falls far short of task-relevant models. Details can be seen in Extended Data Fig. 1.

Based on the above findings, we developed IRES-LM as an ensemble of IRES-UTRLM and IRES-RNAFM. Systematic ablation studies (Fig. 3b) demonstrated that this ensemble strategy yields superior performance compared with either individual model, achieving an AUC of 0.78, AUPR of 0.62 and F1 score of 0.51. This performance gain reflects the complementary strengths of the two models: UTR-LM's specialized knowledge of 5′ UTR regulatory elements and RNA-FM's transferable features from non-coding RNAs that generalize well to 5′ UTR contexts. The ensemble approach also provides enhanced prediction confidence crucial for experimental applications, as consensus predictions between two independently trained models reduce false positive rates and minimize costly validation efforts. Importantly, the predictive performance of IRES-LM remains robust under a homology-controlled, cluster-level cross-validation scheme that prevents highly similar sequences from being shared between training and test sets (Supplementary Information Section E).

For benchmarking, we retrained IRESfinder[11], IRESpy[12] and DeepCIP[13] using the same tenfold cross-validation experimental settings as IRES-LM to ensure consistency. As shown in Fig. 3c, IRES-LM outperformed the three benchmarks, achieving 15% higher AUC and 10% higher AUPR compared with the previous state-of-the-art DeepCIP[13] while improving F1 score by 15% over IRESfinder[11].

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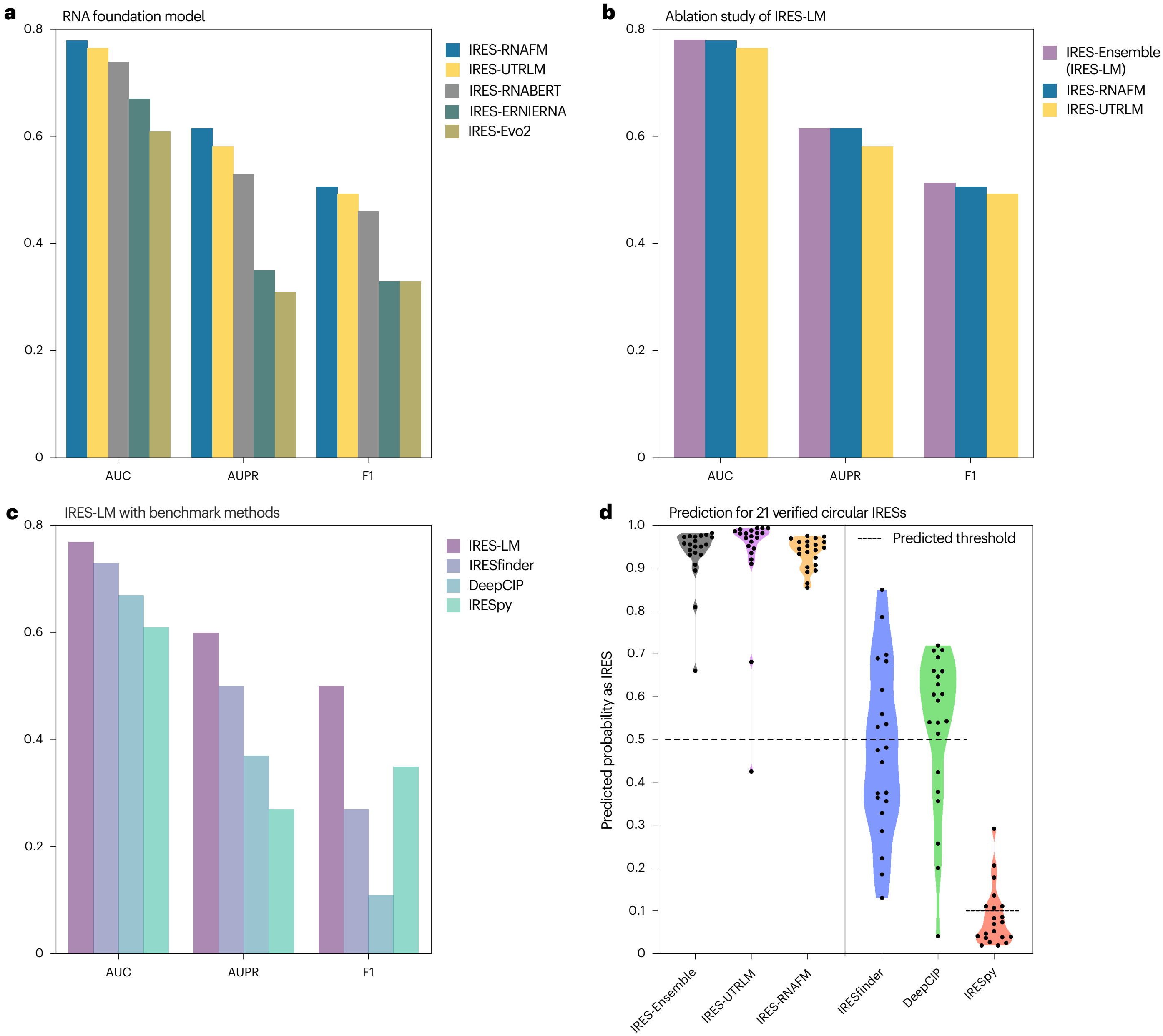


Fig. 3 | IRES-LM more accurately identifies IRES than benchmark methods. **a**, Comparative performance of RNA foundation models (RNA-FM, UTR-LM, RNA-BERT, ERNIE-RNA and Evo-2). **b**, An ablation study of IRES-LM, which is an ensemble of IRES-RNAFM and IRES-UTRLM. **c**, Comparative performance of IRES-LM with benchmark methods (IRESfinder, DeepCIP and IRESpy). **d**, Performance comparison in predicting 21 experimentally validated circRNA IRESs. IRES-LM (that is, IRES-Ensemble) identified all 21 IRESs, whereas IRES-UTRLM mispredicted one. DeepCIP, IRESfinder and IRESpy predicted 15, 9 and 7 IRESs, respectively.

IRES-LM discovers IRESs in experimentally validated circRNAs

To assess model generalization, we evaluated IRES-LM's ability to identify IRESs in circRNAs, despite being trained exclusively on IRESs in linear RNA. This dataset comprises 21 experimentally confirmed IRES sequences from 19 translatable circRNAs, previously documented by DeepCIP[13]. With lengths ranging from 54 to 478 base pairs, these IRES sequences provided a diverse test set for evaluating the robustness of our model to both RNA structure type and sequence length variation.

We benchmarked IRES-LM against three established methods: IRESfinder[11], IRESpy[12] and DeepCIP[13] (Fig. 3d). For threshold selection, we employed 0.5 for IRES-LM and adopted published thresholds of 0.5 for IRESfinder and DeepCIP and 0.1 for IRESpy. By integrating predictions from IRES-UTRLM and IRES-RNAFM components, IRES-LM successfully detected all 21 circRNA IRESs, while its IRES-UTRLM component alone misidentified one sequence. The benchmark methods achieved lower accuracy: DeepCIP[13] detected 15 IRESs, IRESfinder[11] detected 9 and IRESpy[12] detected only 7 of these IRESs.

Beyond experimentally validated IRESs, we extended IRES-LM to analyse the riboCIRC database[23], which contains potentially translatable circRNAs. Our analysis of 487 ribo-circRNAs revealed potential IRES regions in 170 sequences. Detailed analysis of these predictions is available in Supplementary Information Section D.

IRES-LM captures biologically relevant features

To understand how IRES-LM captures biologically relevant features for IRES functionality, we analysed its learned representations using 55,000 sequences with minimum free energy (MFE) values and 5,388 sequences with splicing scores collected by IRESPredictor[24].

Importantly, neither property was included during training, allowing us to assess whether our models spontaneously learned functionally relevant organizational patterns.

Extended Data Fig. 2 visualizes model representations coloured by MFE values and splicing scores through both uniform manifold approximation and projection and principal component analysis. Despite showing distinct clustering patterns in uniform manifold approximation and projection visualizations of the full model representations, both IRES-UTRLM and IRES-RNAFM naturally learned to capture MFE and splicing score variations without explicit training on these properties. Quantitative analysis using principal component analysis confirmed these observations: the first principal component of model representations showed strong correlations with MFE (IRES-UTRLM: Pearson $r = -0.510$; IRES-RNAFM: Pearson $r = 0.570$) and splicing scores (IRES-UTRLM: Pearson $r = -0.410$; IRES-RNAFM: Pearson $r = 0.306$). Although the correlations differ in direction, both models effectively encode these biological features in their learned representations. These results indicate that IRES-LM captures biologically meaningful patterns without explicit supervision, suggesting its potential for discovering novel IRES-related features.

IRES-EA induces functional conversion from non-IRES to IRES

The development of computational tools for rational IRES sequence engineering represents a critical advancement in RNA therapeutics and vaccine design. Although previous studies have explored methods to enhance IRES functionality[8,9], the field has lacked efficient computational approaches that could minimize costly wet-lab experiments while enabling systematic IRES sequence design. To address these limitations, we developed IRES-EA, an evolutionary algorithm that leverages our IRES-LM model to enable mutation-guided exploration of sequence space for inducing or disrupting IRES functionality.

IRES-EA implements an iterative optimization framework to systematically transform initial sequences into targeted functional IRES sequences (Fig. 4a). The algorithm performs sequence optimization through three fundamental steps: sequence masking, mutation sampling and fitness-based selection. In the masking phase, the algorithm masks multiple nucleotide positions within the seed sequence as potential mutation sites. For each masked variant, IRES-LM computes two critical metrics: the nucleotide recovery probability distribution at individual masked positions and the IRES functionality probability of potential recovered sequences. IRES-EA then generates candidate sequences through probabilistic sampling, prioritizing mutations at positions with high recovery probabilities. During the selection phase, candidate sequences with higher predicted IRES probabilities relative to both the initial sequence and current seed variants are selected as seed sequences for subsequent iterations. This optimization process continues until reaching either the predetermined mutation threshold or achieving the desired IRES functionality.

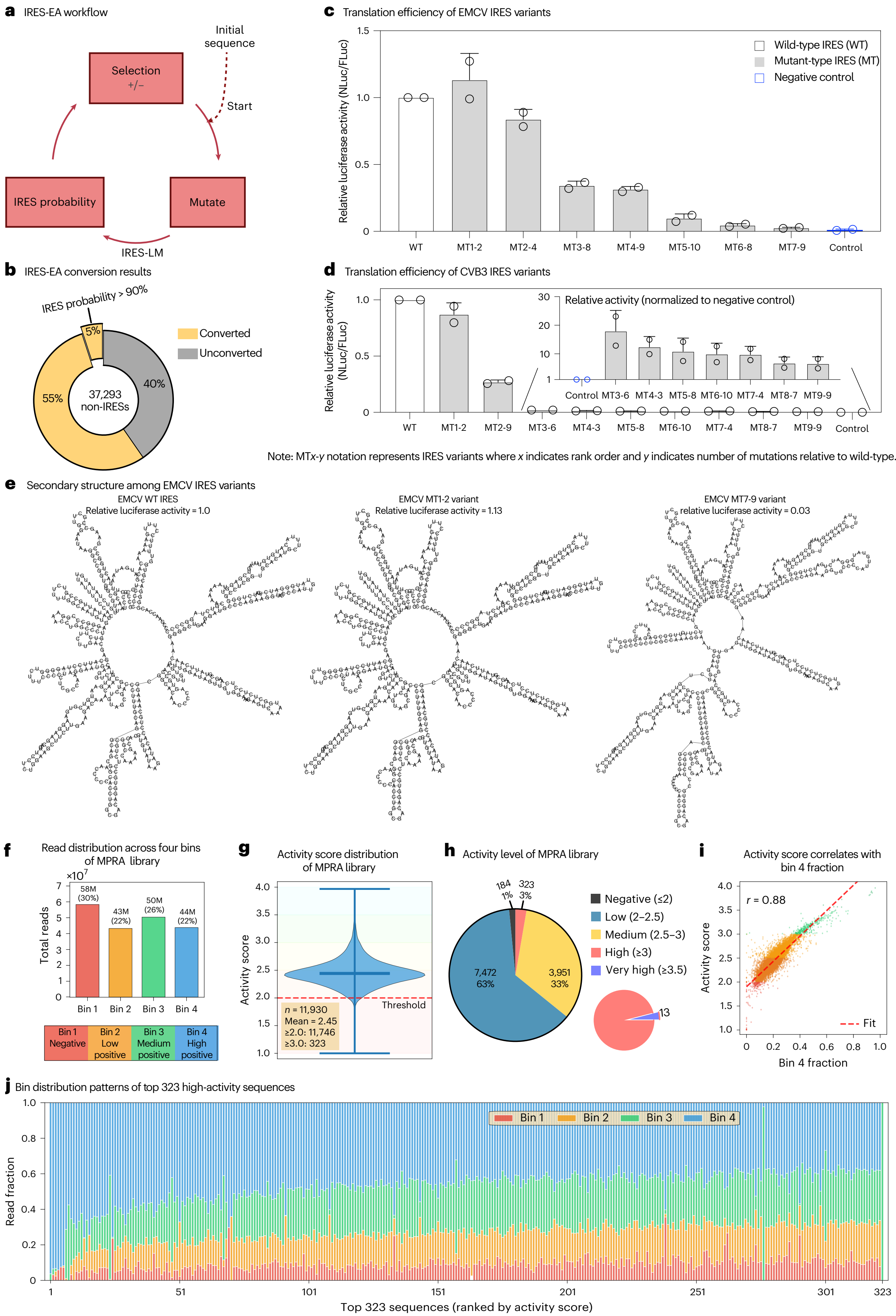
To computationally validate the effectiveness of IRES-EA, we applied it to 37,293 experimentally validated non-IRES sequences of 174-nt length from our training dataset, allowing up to nine mutation sites per sequence. As shown in Fig. 4b, IRES-EA successfully induced predicted functional conversion in 60% (22,243) of these sequences into predicted IRESs (IRES probability >0.5), among which 5% (1,864) achieved high IRES probability exceeding 0.9. Although increasing mutation sites could potentially yield more successful conversions, we limited the mutated number to maintain the original sequence characteristics in this controlled mutational perturbation setting.

To experimentally validate IRES-EA, we selected EMCV IRES (567 nt) and CVB3 IRES (740 nt) as benchmark sequences for directed mutation experiments due to their established high IRES activities and distinct evolutionary origins. Using a bicistronic luciferase reporter assay, we measured IRES activity by normalizing nano luciferase (NLuc) expression driven by IRES sequences to firefly luciferase (FLuc) expression. For each benchmark IRES, we generated sequence variants with up to ten mutation sites predicted to affect IRES activity. The top ten predicted variants of each IRES were experimentally tested and compared with their corresponding wild-type IRES and a negative control. After excluding sequences with experimental anomalies, the results demonstrated that our method can effectively mutate IRESs and identify functional sites within these elements. For EMCV IRES variants (Fig. 4c), one out of seven mutants (MT1-2) achieved higher activity than the wild-type EMCV IRES, and all seven variants maintained substantial IRES activity above the negative control. For CVB3 IRES variants (Fig. 4d), one out of nine mutants (MT1-2) achieved comparable activity to the wild-type CVB3 IRES, and all nine mutated sequences exhibited detectable activity above the negative control. The limited enhancement observed in both IRES families is probably attributed to the inherently high baseline IRES activity of the wild-type sequences. To understand the structural basis for differences in IRES activity, we analysed secondary structures of selected EMCV IRES variants (Fig. 4e). The MT1-2 variant with higher IRES activity maintained a structure similar to wild type, whereas the low-activity MT7-9 variant showed substantial structural differences from the wild type. This structural analysis provides mechanistic insight into how specific mutations affect IRES functionality through alterations in RNA folding. Notably, our dataset for IRES-LM development consisted almost entirely of 174-nt sequences, yet these results validate the generalization and effectiveness of our model in identifying functional elements across different IRES families and sequence lengths.

MPRA experiments validate the IRES-EA efficacy and IRES-LM accuracy. To comprehensively validate IRES-EA beyond the EMCV/CVB3 examples, we performed large-scale MPRA on 12,000 mutated sequences derived from 6,730 negative wild-type sequences. Mutants exhibited substantially elevated predicted IRES probabilities compared with wild types (mean 0.83 versus 0.54), with 60.1% (7,212 sequences) achieving probabilities ≥0.8. The library reflected EA's iterative optimization strategy, comprising 65.5% sequences with six to ten mutations and 34.5% with two to five mutations.

The MPRA workflow employed a dual-fluorescence bicistronic reporter measuring IRES-dependent GFP expression relative to

Fig. 4 | IRES-EA directed mutation and MPRA validation. a, Evolutionary algorithm workflow. Iterative process of sequence masking, IRES-LM guided mutation and fitness-based selection to optimize IRES functionality. **b**, High conversion efficiency. IRES-EA successfully converts 60% of non-IRES sequences ($n = 37{,}293$) to predicted IRES, with 5% achieving probability >90%. **c**, EMCV IRES variant validation. A bicistronic reporter assay shows MT1-2 (two mutations) exhibits higher activity than the wild type (WT); all seven variants maintain activity above negative control. **d**, CVB3 IRES variant validation. MT1-2 (two mutations) achieves activity comparable to the WT; all nine mutants show detectable activity (inset: normalized to negative control). For **c** and **d**, the data are presented as mean ± s.d. from two independent biological replicates ($n = 2$). **e**, Structure–activity relationship among CVB3 IRES variants. High-activity MT1-2 maintains WT-like structure, whereas low-activity MT7-9 shows structural divergence. **f**, Sequencing coverage. Balanced read distribution across bins (~200 million total reads). **g**, High success rate. 98.4% sequences ($n = 11{,}930$ unique library sequences with sufficient barcode read coverage) achieve functional activity (≥2.0, mean 2.45). The central line denotes the mean, and the upper and lower lines indicate the maximum and minimum values, respectively. **h**, Activity level distribution. Five categories showing majority (97%) in the low-to-medium range (2.0–3.0). **i**, Measurement validation. Strong correlation between the bin 4 fraction and activity score ($r = 0.88$) confirms FACS reliability. **j**, Activity ranking confirmation. Top 323 sequences show progressive bin 4 enrichment by rank. For **f**–**j**, the MPRA validation of 12,000 IRES-EA mutated sequences is shown. FACS-based sorting into four bins (bin 1, negative; bins 2–4, positive with increasing activity); activity score from normalized read distributions across bins.


a IRES-EA workflow
Initial sequence
Start
Selection +/−
Mutate
IRES probability
IRES-LM
b IRES-EA conversion results
IRES probability > 90%
5%
Converted
Unconverted
55%
40%
37,293 non-IRESs
c Translation efficiency of EMCV IRES variants
Relative luciferase activity (NLuc/FLuc)
Wild-type IRES (WT)
Mutant-type IRES (MT)
Negative control
WT MT1-2 MT2-4 MT3-8 MT4-9 MT5-10 MT6-8 MT7-9 Control
d Translation efficiency of CVB3 IRES variants
Relative luciferase activity (NLuc/FLuc)
Relative activity (normalized to negative control)
Control MT3-6 MT4-3 MT5-8 MT6-10 MT7-4 MT8-7 MT9-9
WT MT1-2 MT2-9 MT3-6 MT4-3 MT5-8 MT6-10 MT7-4 MT8-7 MT9-9 Control
Note: MTx-y notation represents IRES variants where x indicates rank order and y indicates number of mutations relative to wild-type.
e Secondary structure among EMCV IRES variants
EMCV WT IRES
Relative luciferase activity = 1.0
EMCV MT1-2 variant
Relative luciferase activity = 1.13
EMCV MT7-9 variant
relative luciferase activity = 0.03
f Read distribution across four bins of MPRA library
×10^7
Total reads
58M (30%)
43M (22%)
50M (26%)
44M (22%)
Bin 1 Bin 2 Bin 3 Bin 4
Bin 1 Negative
Bin 2 Low positive
Bin 3 Medium positive
Bin 4 High positive
g Activity score distribution of MPRA library
Activity score
Threshold
n = 11,930
Mean = 2.45
≥2.0: 11,746
≥3.0: 323
h Activity level of MPRA library
Negative (≤2)
Low (2–2.5)
Medium (2.5–3)
High (≥3)
Very high (≥3.5)
184 1%
323 3%
7,472 63%
3,951 33%
13
i Activity score correlates with bin 4 fraction
r = 0.88
Activity score
Fit
Bin 4 fraction
j Bin distribution patterns of top 323 high-activity sequences
Read fraction
Bin 1 Bin 2 Bin 3 Bin 4
Top 323 sequences (ranked by activity score)

constitutive mCherry (Fig. 1). Following lentiviral packaging and transduction, FACS separated cells into four bins based on GFP/mCherry ratios: bin 1 (negative, GFP−) and bins 2–4 (positive with progressively increasing activity, GFP+). High-throughput sequencing achieved ~200 million total reads across bins (Fig. 4f; average 16,408 reads per sequence). Activity score was calculated as weighted averages of normalized read counts[25] to measure the IRES functionality. Activity score ≥2.0 indicates positive IRES activity.

Of 12,000 sequences, 11,930 yielded sufficient coverage, revealing robust IRES functionality with a mean activity score of 2.45 (Fig. 4g). Of these, 98.4% (11,746 sequences) demonstrated positive activity (≥2.0). Activity distribution showed 62.6% (7,472) low (2.0–2.5), 33.1% (3,951) medium (2.5–3.0) and 2.7% (323) high activity (≥3.0) (Fig. 4h). Strong correlation between bin 4 fraction and activity score (r = 0.88, Fig. 4i) confirmed measurement reliability. Analysis of the top 323 high-activity sequences revealed progressive enrichment from bin 1/2 dominance (lower-ranked) to bin 3/4 dominance (higher-ranked), with highest-activity sequences showing near-complete bin 4 enrichment (Fig. 4j), validating both sorting sensitivity and genuine functional diversity among EA-generated sequences.

Collectively, these results validate both IRES-EA's functional induction capability (98.4% of non-IRESs successfully converted to functional IRESs) and IRES-LM's prediction accuracy (98.4% of predicted positives experimentally confirmed), establishing this integrated framework as an effective tool for expanding the functional IRES landscape beyond naturally occurring variants.

IRES-DM de novo designs novel IRESs

De novo design of IRES sequences represents a crucial approach for expanding the library of translation regulatory elements, enabling more precise control of gene expression and the development of novel therapeutic strategies. Although existing computational methods have made progress in IRES sequence engineering, approaches for de novo IRES design remain largely unexplored, with tools such as IRES-EA primarily focusing on modifying natural sequences. To address this limitation, we developed IRES-DM, which implements a DDPM[20] to generate novel IRES sequences. In our approach, IRES-LM guides the generation process through reward-based optimization and sequence screening, enabling the generation of structurally diverse IRES variants that go beyond conventional patterns.

To generate structurally diverse IRES variants, we developed two versions of IRES-DM to accommodate different application scenarios. Variable-length IRES-DM was trained on 9,172 natural IRES sequences (all positive IRES in the dataset) and can generate IRES sequences with lengths ≤200 nt. Fixed-length IRES-DM was specifically trained for 174-nt length using 7,348 174-nt IRES sequences from the dataset (over 80% of the dataset), designed to generate standardized-length IRES sequences. For each length version, we implemented two training strategies: the reward-guided approach leverages IRES-LM as a reward model by calculating cross-entropy loss between predicted and target labels, enabling the model to learn discriminative features that distinguish IRES from non-IRES sequences; the direct-training approach focuses on optimizing sequence generation itself without the reward model component.

To validate the generation capability of variable-length IRES-DM, we selected GenerRNA[21] as the benchmark method—a state-of-the-art RNA generative model that only supports variable-length sequence generation. We fine-tuned GenerRNA on the same 9,172 IRES sequences and had all models (GenerRNA, IRES-DM reward-guided and IRES-DM direct training) generate 9,172 sequences for consistent comparison with the natural IRES training set. Notably, GenerRNA generated sequences contained substantial duplicates, with only 2,944 valid sequences remaining after deduplication, whereas both IRES-DM training strategies produced approximately 6,000 valid sequences. Subsequently, we applied multiple independent quality filtering criteria to the valid sequences, including removal of consecutive repeat nucleotides, IRES probability threshold filtering and length requirements. Variable-length IRES-DM outperformed GenerRNA across all quality criteria (Fig. 5a). For fixed-length IRES-DM, since GenerRNA cannot generate fixed-length sequences, we used random sequence generation as the benchmark, generating 7,348 sequences to match the training scale of the fixed-length model. Fixed-length IRES-DM demonstrated the ability to learn IRES-specific patterns rather than random patterns (Fig. 5b). IRES-DM outperformed random generation across all quality filters. Particularly in the IRES probability >0.5 filtering, IRES-DM retained approximately 2,100 sequences while random sequences retained only 952. These results confirm that IRES-DM effectively captures the key features of natural IRES sequences rather than simply generating random patterns. Moreover, the two training strategies (reward-guided and direct training) showed similar performance, indicating that IRES-DM is robust and does not depend on specific training strategies.

To validate the capability of IRES-DM, we experimentally tested two generated sequences that closely resembled the VCIP IRES (95 nt), with the first sequence (generated-1) produced by our reward-guided approach and the second (generated-2) by our direct-training approach, differing by only eight and seven nucleotides from the natural IRES, respectively. As shown in Fig. 5c, generated-1 exhibited obvious IRES activity compared with the negative control. The fact that our model converged on sequences similar to those found in nature while maintaining functional activity suggests that IRES-DM has successfully captured the essential sequence–function relationships governing IRES elements.

Furthermore, IRES-DM demonstrated the ability to generate functional sequences through diverse evolutionary paths. Figure 5d shows an exciting example where our generated sequence maintains similar secondary structural features to the natural BiP IRES[22] despite sharing only 27.6% sequence similarity. It indicates that IRES-DM can preserve

Fig. 5 | Comprehensive evaluation of IRES-DM generated sequences. **a**, Variable-length IRES-DM outperforms GenerRNA. Generates 9,172 high-quality sequences with superior performance across all metrics. **b**, Fixed-length IRES-DM outperforms random generation. Generates 7,348 sequences with higher yield of high-probability candidates. **c**, Experimental validation of generated IRES activity. Two sequences resembling VCIP IRES show functional activity above negative control. Data are presented as mean ± s.d. from two independent biological replicates (n = 2). **d**, Structural conservation with sequence divergence. Generated sequence maintains BiP IRES structure despite only 27.6% sequence identity. **e**, Motif composition analysis. Generated sequences (n = 1,168) recapitulate natural k-mer patterns at both 3-mer and 6-mer lengths. **f**–**l**, MPRA validation of 12,000 IRES-DM generated sequences. FACS-based sorting into four bins (bin 1, negative; bins 2–4, positive with increasing activity); activity score from normalized read distributions across bins. **f**, Sequencing coverage. Balanced distribution across bins with ~200 million total reads. **g**, High functional success rate. A total of 99.3% of sequences (n = 11,887 unique library sequences with sufficient barcode read coverage) achieve IRES activity ≥2.0 with mean 2.50. **h**, Model type comparison. Both 174-nt fixed and variable-length models show similar activity profiles with ~99% success rates. **i**, Length–activity relationship. Low-activity sequences exhibit broader length distributions with enrichment of longer sequences (approaching 200 nt). **j**, Activity across length categories. Similar mean scores (2.43–2.50) across categories, but <174-nt sequences show constrained maximum activity (~2.7). **k**, Training strategy equivalence. Direct-training and reward-guided approaches achieve similar performance for both fix-length and variable-length models (>98% success). **l**, Activity ranking validation. The top 196 sequences show progressive bin 4 enrichment correlating with activity rank. For **g**–**k**, n represents the number of unique sequences in each category; for violin plots, the central line denotes the mean, and the upper and lower lines indicate the maximum and minimum values, respectively.

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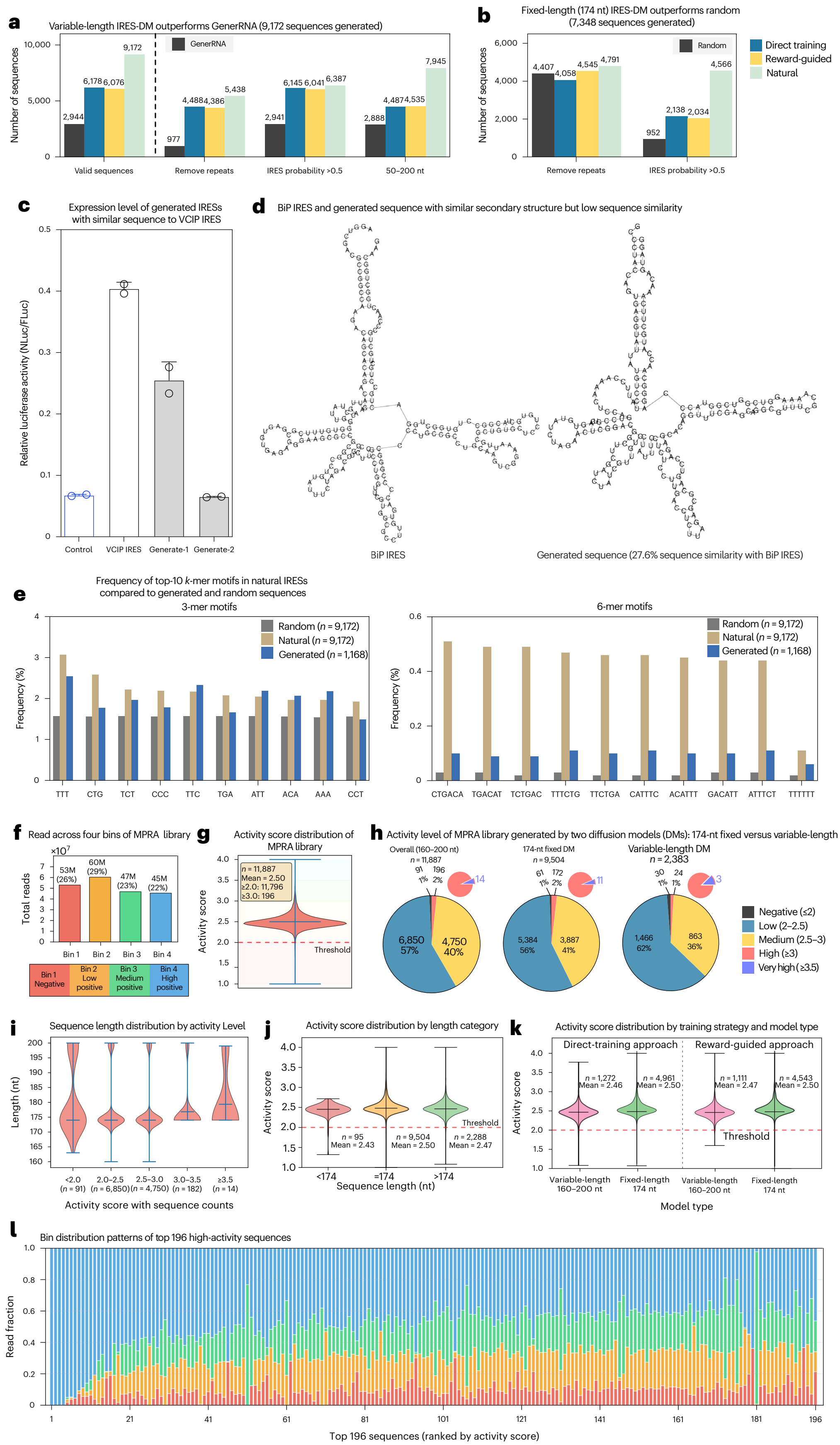

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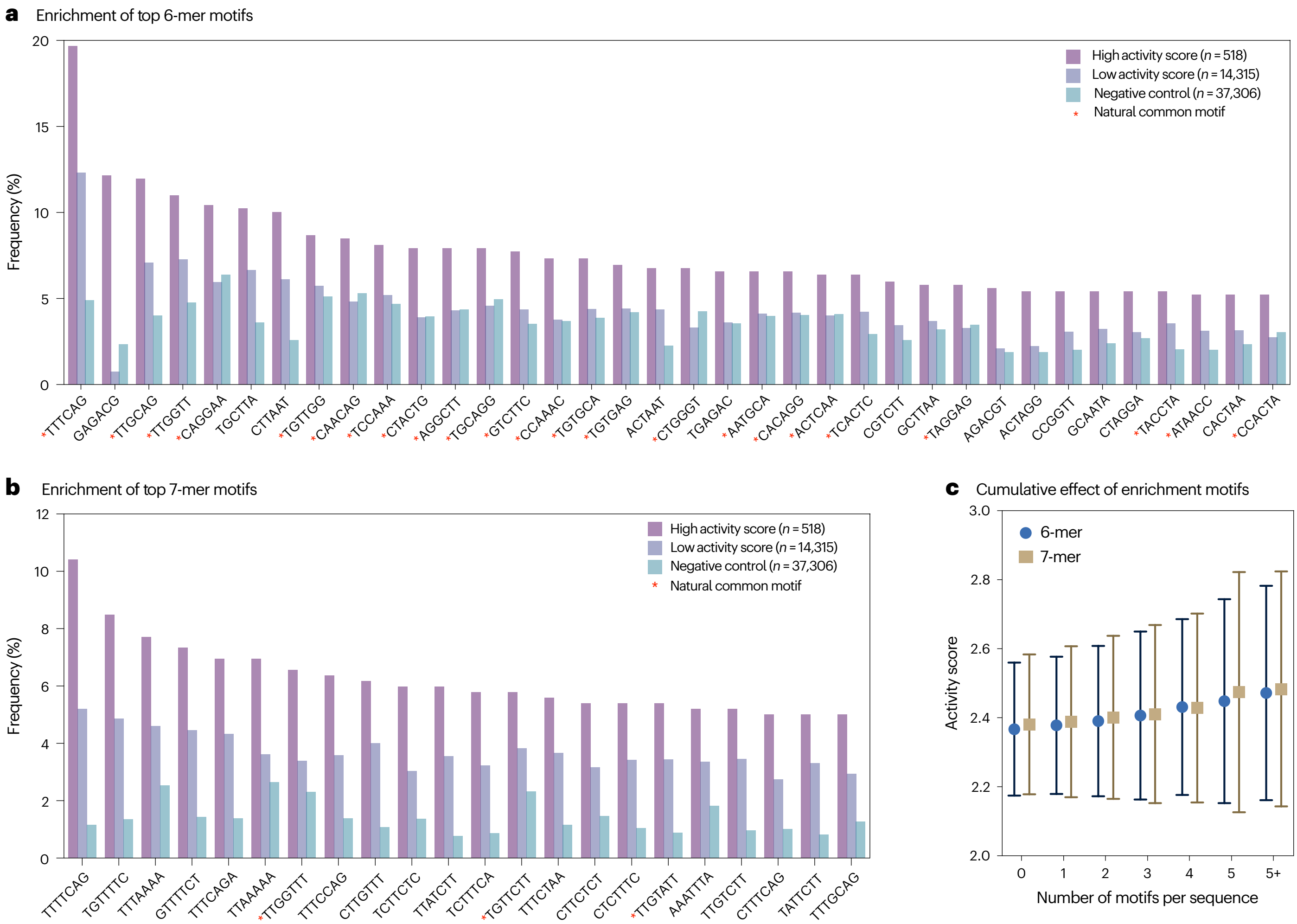

Fig. 6 | Enrichment and cumulative effects of high-activity related motifs on IRES activity. a,**b**, Frequency (in percentage) of high-activity related motifs across high-activity, low-activity and negative control groups. The red asterisks mark natural-prevalent motifs (also common in natural IRES sequences); unmarked motifs are design-enriched (rare in natural IRES but enriched in artificial intelligence-generated high-activity sequences). **a**, 6-mer motifs. **b**, 7-mer motifs. **c**, Activity score versus motif count per sequence for 6-mer and 7-mer motifs, showing positive correlation between motif abundance and IRES activity. For each motif-count group, *n* represents the number of unique sequences within that group (ranging from $n = 549$ to 7,453 for 6-mers and from $n = 100$ to 11,480 for 7-mers). Data are presented as mean ± s.d. calculated across sequences within each group.

essential structural elements while exploring diverse sequence space, suggesting multiple viable strategies for IRES design.

We further analysed the motif composition of our generated sequences compared with natural and random sequences (created by sampling nucleotides with equal probability). We selected the top ten motifs from natural IRESs based on frequency and calculated their occurrence across all sequence sets. As shown in Fig. 5e, the generated sequences closely match the motif distribution patterns of natural sequences, even as the motif length increases from $k = 3$ (short motifs) to $k = 6$ (long motifs). By contrast, the differences between random and natural sequences become more pronounced with longer motifs. These findings indicate that IRES-DM effectively captures the k-mer composition characteristics of natural IRES sequences, further validating its capability to learn fundamental sequence patterns important for IRES functionality. To comprehensively validate the functional capacity of IRES-DM generated sequences, we performed large-scale MPRA experiments.

MPRA experiments validate the IRES-DM efficacy and IRES-LM accuracy. To comprehensively validate IRES-DM's generative capacity beyond individual examples, we constructed a 12,000-sequence MPRA library from four generation strategies: 174-nt fixed-length models (direct training: $n = 5{,}005$; reward guided: $n = 4{,}577$) and variable-length models (direct training: $n = 1{,}293$; reward guided: $n = 1{,}125$; length 160–200 nt). The library showed favourable computational characteristics (mean IRES predicted probability 0.84, 24.4% >0.9). FACS-seq analysis recovered 11,887 sequences (98.9%) with ~200 million total reads (Fig. 5f). Remarkably, 99.3% (11,796/11,887) exhibited functional IRES activity (score ≥2.0, mean 2.50) (Fig. 5g), validating both IRES-DM's generative capability and IRES-LM's predictive accuracy.

Comparative analysis between model architectures revealed notable insights (Fig. 5h). The 174 nt fixed model contributed most sequences ($n = 9{,}504$, 80%) because the training dataset mainly comprises 174-nt sequences. Despite this bias, 99% of variable-length sequences ($n = 2{,}383$, 20%) were measured as IRES by experiments, demonstrating IRES-LM's generalization capability beyond its primary training length. However, analysis of sequence length distribution across activity levels (Fig. 5i,j) revealed variable-length model limitations: lower-activity sequences (<2.0, $n = 91$) showed enrichment of longer sequences, whereas shorter sequences (<174 nt, $n = 95$) failed to achieve high activity (max ~2.7). Moreover, evaluation of training strategies showed minimal differences (Fig. 5k): both variable-length (direct: mean 2.46, n = 1,272; reward-guided: mean 2.47, n = 1,111) and 174-nt models (direct/reward-guided: mean 2.50; p = 0.421) achieved similar performance, highlighting IRES-DM's robustness regardless of training strategy. In addition, analysis of top 196 high-activity sequences revealed progressive enrichment from bin 1/2 dominance (lower-ranked) to bin 3/4 dominance (higher-ranked) (Fig. 5l).

These MPRA results provide strong validation for IRES-DM. Combined with the computational analyses showing superior performance over GenerRNA, structural diversity exemplified by the BiP-like sequence, and capture of natural motif patterns, the 99.3% experimental success rate across 11,887 sequences confirms IRES-DM's capacity for de novo generation of diverse, functional IRES elements. This demonstrates IRES-DM as a powerful approach for expanding the IRES repertoire for applications in synthetic biology and gene therapy.

High-activity related *k*-mer motifs show activity-dependent enrichment and cumulative effects

We performed systematic analysis on 6-mers and 7-mers. Shorter *k*-mers (3-mer, 4-mer and 5-mer), despite their high occurrence frequency, lack sufficient sequence specificity to capture meaningful functional motifs; whereas longer *k*-mers (8-mer and above) suffer from insufficient statistical power due to combinatorial explosion and excessively low individual motif frequencies. Therefore, we selected 6-mers and 7-mers as analysis targets, achieving a balance between sequence specificity and statistical power.

Through enrichment analysis, we identified 36 high-activity related 6-mer motifs and 22 high-activity related 7-mer motifs (Fig. 6a,b). For 6-mers, among the 36 motifs, 13 were design-enriched (frequency ≤5% in natural IRES and high/natural frequency ratio ≥2), and 23 were natural-prevalent (also common in natural IRES). For 7-mers, among the 22 motifs, 19 were design-enriched and 3 were natural-prevalent. These motifs satisfied stringent filtering criteria: high group frequency exceeding both low and negative control groups, high/negative enrichment ≥1.5 ($P \leq 0.05$) and high/low gradient ≥1.5, clearly demonstrating activity-dependent characteristics. These findings reveal that the proposed model not only explored high-activity sequence space underutilized by natural evolution (design-enriched motifs) but also successfully learned known, universal IRES functional features (natural-prevalent motifs).

Further cumulative effect analysis revealed a positive correlation between the number of high-activity related motifs per sequence and IRES activity (Fig. 6c). For both 6-mers and 7-mers, mean activity score progressively increased with motif count, with sequences containing no motifs showing the lowest average activity and those containing five or more motifs showing the highest. This trend suggests that multiple motifs may have additive or synergistic effects, supporting that these identified motifs are indeed functional determinants of high activity and that their combination can effectively enhance IRES activity.

Discussion

The IRES framework presented in this study represents an important advance in the field of RNA translation regulation through three integrated components: IRES-LM for identification, IRES-EA for mutation-guided functional induction and IRES-DM for de novo generation. Our IRES-LM model demonstrates superior predictive capabilities, achieving a 15% improvement in AUC and F1 scores compared with existing methods. By systematically evaluating multiple RNA foundation models and selecting task-aligned architectures, IRES-LM captures biologically meaningful RNA features, including secondary structure stability and splicing signals, without explicit training on these properties. Beyond excelling in linear mRNA IRES prediction, IRES-LM exhibits remarkable cross-applicability by correctly identifying all 21 experimentally validated circRNA IRESs, demonstrating that learned representations generalize across RNA structure types and revealing universal patterns underlying IRES functionality.

Building upon this robust identification foundation, IRES-EA successfully addresses the challenge of systematic modelling of mutation-induced functional state transitions. The algorithm's effectiveness is evidenced by its ability to convert 60% of 37,293 non-IRES sequences into predicted functional IRESs, with 5% achieving high IRES probability exceeding 90%. Large-scale MPRA validation of 12,000 mutated sequences confirms these computational predictions with exceptional fidelity: 98.4% of predicted IRESs demonstrated functional activity experimentally, establishing both IRES-EA's functional induction capability and IRES-LM's prediction accuracy. Targeted validation on phylogenetically distinct EMCV and CVB3 IRES families further revealed that high-activity mutants maintain wild-type-like secondary structures, providing a structural basis for understanding sequence–function relationships in IRES engineering.

The IRES-DM component complements the framework by enabling de novo generation of functional IRES sequences. Benchmarking against GenerRNA, a state-of-the-art RNA generative model, demonstrated IRES-DM's superior performance in generating high-quality, non-duplicated sequences, while comparison with random sequence generation confirmed genuine learning of IRES-specific patterns. Large-scale MPRA validation of 12,000 generated sequences achieved 99.3% functional success rate, with consistent performance across both fixed-length (174 nt) and variable-length (≤200 nt) model versions and training strategies, confirming robust generative capability. Motif enrichment analysis revealed that IRES-DM not only captures natural-prevalent high-activity motifs but also discovers design-enriched motifs rarely found in natural sequences, demonstrating exploration of novel sequence space beyond natural evolution. Moreover, IRES-DM generates sequences sharing only 27.6% sequence similarity with natural IRESs such as BiP IRES while maintaining remarkably similar secondary structures, revealing multiple viable evolutionary paths for achieving IRES functionality.

Future work will focus on expanding the capabilities of this framework in several key directions. First, incorporating structural information into both IRES-EA and IRES-DM would enable more precise control over RNA folding patterns, potentially addressing current limitations in variable-length sequence generation and expanding design space for circRNA IRESs with their unique structural constraints. Second, systematic investigation of the identified design-enriched motifs could reveal novel sequence determinants of high IRES activity, guiding rational design of enhanced IRES elements. Third, extending the framework to generate IRES variants optimized for specific cellular contexts or stress conditions would enhance therapeutic applicability, particularly for mRNA and circRNA vaccines where translational efficiency under diverse physiological states is critical. These advancements would further enhance the practical utility of our framework for advancing RNA-based therapeutics and synthetic biology applications.

Methods

Datasets

We collected a dataset of binary-labelled sequences indicative of their functionality as IRES, retrieved from multiple data sources, including the study of Weingarten-Gabbay et al.[16], IRESbase[17], IRESite[18], Rfam[19] and the training set of IRESpred[10]. These records were filtered by sequence to remove duplication between different data sources. The final dataset contains 46,774 records, with 9,172 IRES sequences and 37,602 non-IRES sequences. In the dataset, the longest sequence is 1731 nt, with over 80% of sequences being 174 nt long.

IRES-LM: IRES prediction by consensus language models

Given that IRES elements act as translational regulatory factors in the 5′ UTR, we transferred pretrained language models UTR-LM and RNA-FM to predict the IRES annotation for RNA sequences. UTR-LM is pretrained on 5′ UTR sequences, and RNA-FM treats 5′ UTR prediction as a downstream task. Both models achieved satisfactory performance on 5′ UTR-related prediction. We concatenated the language model of UTR-LM and RNA-FM with a simple MLP as the IRES classifier. For differentiation, we named them IRES-UTRLM and IRES-RNAFM, respectively, and collectively referred to them as IRES-LM. The workflow is shown in Fig. 2a.

The architecture of IRES-UTRLM consists of a six-layer transformer block as the pretrained language model, with a pretrained embedding size of 128. The architecture of IRES-RNAFM consists of a 12-layer transformer block as the pretrained language model, with a pretrained embedding size of 640. For both models, a simple MLP with a 64-neuron feed-forward layer and a 2-neuron output layer is added to <CLS> token and used for IRES classification. All parameters were fine-tuned on 46,774 binary-labelled sequences.

IRES-LM has two primary training objectives. First, to learn the IRES representation within the language model, we minimized the cross-entropy loss for recovering masked nucleotides in the masked language model. The loss function is $L_{\mathrm{MLM}} = -\Sigma_{i\in\mathrm{masked}} \log P(\mathrm{token}_i|\mathrm{context})$. Second, to predict whether an input sequence is IRES or not, we minimized the cross-entropy loss $L_{\mathrm{classification}} = -\Sigma_{x\in X}(y_x \log \hat{y}_x + (1-y_x)\log(1-\hat{y}_x))$ through back-propagation, where y_x and $\hat{y}_x$ stand for the ground-truth label and the predicted probability of the label being IRES for the sequence x, respectively.

Implementation and evaluation. For implementation, IRES-UTRLM used a classification loss weight of 20.0 and a masked language model loss weight of 1.0, with a masking probability of 0.15. IRES-RNAFM used a classification loss weight of 2.0 and a masked language model loss weight of 1.0, with a masking probability of 0.15, with inputs truncated to a maximum length of 1,024 tokens. IRES-UTRLM was trained for 100 epochs with a batch token size of 2,048, IRES-RNAFM was trained for 10 epochs with a larger batch token size of 4,096. Both models were fine-tuned with Adam at a learning rate of 1×10^{-4}.

All models were evaluated under identical tenfold stratified splits using the same random seed. Training was conducted on 32-GB NVIDIA Tesla graphics processing unit (GPUs), with IRES-UTRLM requiring approximately 355 s per epoch across fouor GPUs, and IRES-RNAFM required approximately 828 s per epoch. Model performance was quantified using AUC, AUPR and F1 score to evaluate classification performance under data imbalance.

IRES-EA: mutation-guided functional state induction by evolutionary algorithm

The IRES-EA algorithm aims to achieve two primary goals in IRES mutation effect modelling: (1) mutate non-IRES sequences to IRES sequences to enhance protein production of vaccines and (2) mutate IRES sequences to non-IRES sequences to decrease protein production of viruses. Inspired by the ability of language models to improve sequence characteristics through masked language model[26], we developed an evolutionary algorithm based on IRES-LM for directed mutations. The workflow is shown in Fig. 2b.

The process begins by masking nucleotide sites on the wild-type sequence to generate a series of masked sequences. For each sequence, IRES-LM predicts IRES probabilities and recovery probabilities for each masked nucleotide. The nucleotide with the highest recovery probability is selected as the predicted recovery sequence, and additional mutated sequences are sampled based on these probabilities. These sequences are then filtered and sampled based on their predicted IRES annotation probabilities compared with the wild-type and seed sequences. The filtered sequences are used as seeds for the next iteration of mutations, repeating the above process until the desired number of mutation sites or mutated sequences is achieved.

IRES-EA supports configurations including specifying the mutation region of the seed sequence, the number of masked nucleotides per sequence and the maximum number of mutations per sequence. The final mutated sequences are sorted and compared with the wild-type sequence.

Masking strategy. Initially, the nucleotide sites for masking were selected randomly. The sequence is segmented into three parts: the region before the mutation, the target mutation region and the region after the mutation. For random replacement, a specified number of positions within the target segment are selected randomly and masked. Alternatively, for continuous replacement, a random starting position is chosen, and a specified number of consecutive positions are masked. The modified target segment is then reassembled with the presegment and postsegment to form the final masked sequence.

To enhance this strategy, we introduced an attention-based masking approach based on IRES-LM model. The attention weights for the target region are extracted from the model. Positions within the target segment are selected based on the highest attention weights, masking the top-scoring sites. Alternatively, the target region is scanned to find the subsequence that has the highest average attention weight, and this continuous segment is masked. This strategy targets nucleotides most relevant to IRES annotation, potentially improving mutation efficiency.

Recovery strategy. The recovery process involves reconstructing masked nucleotide sites using the IRES-LM model's predictions. For each masked sequence, multiple mutated sequences are generated. If a nucleotide site is masked, the recovery probabilities of nucleotide bases (A, G, C and U) are calculated from IRES-LM. The recovery sequence is initially constructed by replacing the masked site with the most probable token. Additional mutated sequences are generated by sampling tokens according to their recovery probabilities and replacing the masked sites iteratively. This enables diverse mutated sequences by considering multiple potential recoveries for each masked site.

We also support custom nucleotide substitution rules. For example, substitutions can be restricted (for example, A to G) to meet gene-editing constraints. Within the target region, positions with specific nucleotides are identified and replaced according to the rule. This enables controlled mutations that satisfy predefined editing requirements.

IRES-DM: generate new IRES by conditioned denoising diffusion model

The DDPM[20] transforms noise from a simple distribution into a meaningful data sample, using a neural network that learns to gradually remove noise from the data. For RNA sequences, this involves two processes over a finite number of time steps T:

(1) A predefined forward diffusion process q that first converts an RNA sequence into a one-hot encoded latent vector, then gradually adds Gaussian noise to this vector until only noise remains. This process is implemented by taking a random sample S_0 from the real sequence distribution $q(S_0)$. uniformly sampling a noise level t uniformly between 1 and T (that is, a random time step), and then sampling noise from a Gaussian distribution to add to the sequence at level t, progressively corrupting the sequence.

(2) A learned reverse denoising process p_θ involves training a neural network to gradually remove noise, transforming pure noise back into a real RNA sequence. The U-Net is trained to predict the noise based on the corrupted sequence S_t, which is the noise applied to S_0 according to the known schedule β_t.

The training algorithm is detailed as follows, and shown in Fig. 2c.

The forward diffusion process. The forward diffusion process q gradually adds noise to a sequence from the real distribution over several time steps T, according to a variance schedule. The variance schedule

β_t is precomputed by a linear scheduler, where the noise variance β_t increases linearly from 0.0001 to 0.2 over $T = 50$ time steps.

Starting at $t = 0$, a real RNA sequence S_0 is sampled from the real sequence distribution $q(S_0)$. The forward process first converts this sequence into a one-hot encoded latent vector. Then, at each time step t, Gaussian noise is added to the sequence and obtained the corrupted sequence S_t. This forward diffusion process follows the rule $q(S_t|S_{t-1}) = N(S_t; \sqrt{1-\beta_t}S_{t-1}, \beta_t I)$. According to the independent additivity of Gaussian distribution, S_t can be obtained conditioned on S_0 without repeatedly t times applying q. This forward diffusion process q follows $q(S_t|S_0) = N(S_t; \sqrt{\overline{\alpha_t}}S_0, (1-\alpha_t)I)$ where $\alpha_t := \sqrt{1-\beta_t}$ and $\overline{\alpha_t} := \Pi_{i=1}^{t}\alpha_i$. Thus, each corrupted sequence S_t at time step t is drawn from a conditional Gaussian distribution with $\mu_t = \sqrt{\overline{\alpha_t}}S_0$ and $\sigma_t^2 = (1-\alpha_t)$, which we can do by sampling $\epsilon \approx N(0, I)$ and then setting $S_t = \sqrt{\overline{\alpha_t}}S_0 + \sqrt{(1-\alpha_t)}\epsilon$. Note that S_T is pure Gaussian noise if the schedule is set appropriately.

Reverse diffusion process via noise predictor U-Net. We developed a diffusion model capable of conditionally generating both IRES and non-IRES sequences. Thus, the reverse process p involves gradually 'denoising' a random Gaussian noise S_T to recover a sequence S_0 from the real distribution using the conditional distribution $p(S_{t-1}|S_t, y)$, where $y = 0$ represents the sequence S_0 is non-IRES, and $y = 1$ represents the sequence S_0 is IRES. However, $p(S_{t-1}|S_t, y)$ is intractable because it requires knowledge of all possible sequence distributions. We leverage a neural network to approximate the conditional probability distribution $p_\theta(S_{t-1}|S_t, y)$, where θ represents the neural network parameters, which are updated via gradient descent.

We assume the reverse process is Gaussian and parametrize it as $p_\theta(S_{t-1}|S_t, y) = N(S_{t-1}; \mu_\theta(S_t, t, y), \beta_t I)$, where the mean and variance are conditioned on the noise level t. Following DDPM, we keep the variance fixed at $\beta_t I$ and only learn the mean μ_θ. The mean is computed as $\mu_\theta(S_t, t, y) = \frac{1}{\sqrt{\alpha_t}}(S_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha_t}}}\epsilon_\theta(S_t, t, y))$, where $\epsilon_\theta(S_t, t, y)$ represents the added noise at noise level t. Thus, we constructed the neural network as a noise predictor to predict $\epsilon_\theta(S_t, t, y)$.

The neural network is optimized using three loss functions: mean absolute error (MAE) loss, negative log-likelihood (NLL) loss, and cross-entropy (CE) loss.

The MAE loss measures the difference between the true and predicted Gaussian noise. For a random time step t given $\epsilon \approx N(0, I)$, the objective function for the MAE loss is formulated as $L_{\mathrm{MAE}} = |\epsilon - \epsilon_\theta(S_t, t, y)| = |\epsilon - \epsilon_\theta(\sqrt{\overline{\alpha_t}}S_0 + \sqrt{1-\overline{\alpha_t}}\epsilon, t, y)|$. Here S_0 is the initial real sequence, the noise level t is determined by the fixed forward process, ϵ is the pure noise sampled at time step t and $\epsilon_\theta(S_t, t, y)$ is the neural network's prediction of the noise at time step t given the noisy sequence S_t.

Given the predicted noise $\epsilon_\theta(S_t, t, y)$, the noisy sequence S_t and the time step t, the inferred real sequence $\widehat{S_0}(S_t, t, y)$ can be computed using the formula $\widehat{S_0}(S_t, t, y) = \frac{1}{\sqrt{\overline{\alpha_t}}}S_t - \frac{1}{\sqrt{\overline{\alpha_t}-1}}\epsilon_\theta(S_t, t, y)$.

Based on $\widehat{S_0}(S_t, t, y)$, we define two additional objective functions:

(1) The NLL loss quantifies how well the diffusion model can predict the original sequence $\widehat{S_0}(S_t, t, y)$ from the noisy intermediate sequence S_t at a given time step. It is formulated as $L_{\mathrm{NLL}} = -\log p(S_0|\widehat{S_0}(S_t, t, y))$, where $\widehat{S_0}(S_t, t, y) = \frac{1}{\sqrt{\overline{\alpha_t}}}S_t - \frac{1}{\sqrt{\overline{\alpha_t}-1}}\epsilon_\theta(S_t, t, y)$.

(2) The CE loss measures the discrepancy between the ground-true label y of S_0 and the probability p that predicting $\widehat{S_0}(S_t, t, y)$ as IRES. It is formulated as $L_{\mathrm{CE}} = -[y\log p + (1-y)\log(1-p)]$.

These loss functions collectively guide the neural network to effectively learn the noise prediction, improving its ability to denoise sequences for the direction of specific IRES annotation.

We employed U-Net[27] with ResNets[28] for noise prediction (Fig. 2c). U-Net encodes the input sequence with conditions into a compact representation and then decodes it back to the original sequence size, retaining essential features. Inspired by DDPM[20], we integrated multihead self-attention[29] and multihead linear attention[30] between the convolutional blocks.

The U-Net processes the noisy sequence S_t with the respective IRES annotation l and the time step t and outputs the predicted noise $\epsilon_\theta(S_t, t, l)$. Initially, a convolutional layer is applied to the noisy sequence S_t. As the parameters of the U-Net are shared across time steps and IRES annotations, we employ sinusoidal position embedding[30] to encode the time step t and word embeddings for the IRES annotation l. Both embeddings, with a dimensionality of 800, are added to subsequent ResNet blocks. The network then proceeds through three downsampling stages, each consisting of two ResNet blocks, linear attention and a downsampling operation. In the middle of the network, additional two ResNet blocks interleaved with self-attention are applied. Next, three upsampling stages follow, each consisting of two ResNet blocks, linear attention and an upsample operation. Finally, a ResNet block followed by a convolutional layer completes the process of predicting the noise.

Training IRES-DM and generating new sequences. In the IRES dataset, the longest sequence is 1,731 nt, with over 80% of sequences being 174-nt long. Based on this length distribution, we developed two versions of IRES-DM. (1) For the variable-length model, we trained on all 9,172 positive IRES sequences. Sequences were standardized to 200 nt: those >200 nt were truncated to the rightmost 200 nt, whereas shorter sequences were padded with the <PAD> token, with an <EOS> token inserted before padding. (2) For the fixed-length 174-nt model, we used 7,348 sequences of exactly 174 nt from the dataset. For both versions, these sequences were one-hot encoded. During training, the <PAD> tokens were masked when calculating the loss function and noise. Class weights were applied to handle class imbalance.

For model training, we developed two training strategies: a reward-guided approach that leverages IRES-LM as a reward model by calculating cross-entropy loss between predicted and target labels and a direct-training approach that focuses on optimizing sequence generation without the reward model component for greater computational efficiency and potential sequence diversity. These strategies were applied to both the variable-length model (trained on 9,172 sequences) and the fixed-length model (trained on 7,348 sequences).

Once the diffusion model has been trained, we can generate new RNA sequences conditioned on noise and IRES annotation. The process involves reversing the diffusion process, starting from pure noise sampled from a Gaussian distribution at the time step $t = T$, and gradually denoising it to recover a one-hot encoded sequence from the real distribution at the time step $t = 0$. This is done using the trained U-Net model, which predicts the noise to be subtracted at each step.

The model was trained for 150 epochs on a single 32-GB NVIDIA Tesla GPU. The reward-guided approach, which leveraged the IRES-LM classifier for reward calculation, required approximately 352 s per epoch, whereas the direct-training approach needed only 225 s per epoch.

IRES screen

Cell culture. HEK293T cells (Sigma-Aldrich, cat. no. 12022001-1VL) were maintained in Dulbecco's modified Eagle's medium (Gibco) supplemented with 10% fetal bovine serum and 1% penicillin–streptomycin (Thermo Fisher Scientific).

IRES PCR amplification. To generate the desired insert for the IRES assay, we utilized PCR amplification with Phanta Flash Master Mix (Vazyme Biotech, P510). Primers were designed to create the necessary overlaps between the insert and the vector and diluted to a stock concentration of 100 μM. PCR and thermocycling conditions were performed according to the manufacturer's instructions for Phanta Flash Super-Fidelity DNA Polymerase.

Reporter construction. The reporter mCherry–IRES–eGFP reporter construct (mCherry–IRES–eGFP) comprised an mCherry-T2A-puromycin resistance cassette (PuroR) driven by an independent promoter and terminated by its own polyadenylation signal. A cloning cassette consisting of two inward-facing BsmBI sites was placed downstream of the mCherry-T2A-PuroR module to accept IRES inserts. An eGFP open reading frame lacking its own promoter was positioned 3′ to the BsmBI cassette such that eGFP expression depended on the inserted IRES. The entire construct was assembled in a lentiviral transfer plasmid for subsequent library packaging. Requests for plasmid should be addressed to the corresponding author.

Gel extraction. PCR products were subjected to gel electrophoresis to confirm the correct product size. Bands corresponding to the desired amplicons were excised from the gel and purified. This step ensured high purity of the PCR products, eliminating any non-specific amplification.

Transformation. IRES library cloning and bacterial transformation were performed essentially as described previously[31]. The IRES oligonucleotide library (GenScript) was PCR-amplified and cloned into the backbone plasmid by NEBridge Golden Gate Assembly Kit (E1602L). The assembly product was electroporated into ElectroMAX Stbl4 Competent Cells (Thermo Fisher, 11635018) according to the manufacturer's instructions. Transformant colonies were scraped in bulk to preserve library complexity, and plasmid DNA was purified using a Plasmid Plus Maxi Kit (Qiagen).

Lentivirus production and infection. Lentivirus production was performed as described previously[32,33], and HEK293T cells were seeded in 15-cm dishes at 1.3×10^7 cells per dish 24 h before transfection. Linear polyethylenimine (PEI; 40 kDa, Kyfora Bio) was used at a 4:1 (wt/wt) PEI:DNA ratio. To produce the lentiviral IRES library, cells in each 15-cm dish were transfected with 3 μg pMD2.G (Addgene #12259), 12 μg pCMV-ΔR8.2 (Addgene #12263) and 12 μg IRES library plasmid using PEI. The culture medium was replaced 4 h after transfection. Viral supernatants were collected at 24 h and 48 h post transfection. Viral stocks were titrated by serial-dilution infection of HEK293T cells. For IRES library transductions, 3.6×10^8 cells were infected at a multiplicity of infection (MOI) of 0.1 per library and subjected to puromycin (1 ng μl^{-1}) selection for 5 days before flow-cytometric sorting.

FACS sorting. Cells were sorted on a FACSAria II cell sorter (BD Biosciences). MCherry fluorescence was used to normalize infection efficiency. mCherry+/eGFP+ cells were partitioned into four bins according to mCherry-normalized eGFP signal intensity (eGFP/mCherry). For each library, approximately 3.6×10^6 cells were collected.

Next-generation sequencing. Total DNA from each sorted expression bin was prepared by lysis with QuickExtract DNA Extraction Solution (Lucigen). All DNA used as PCR template. Sequencing libraries were generated by two rounds of PCR using Phanta Flash Super-Fidelity DNA Polymerase (Vazyme, P521). PCR products were resolved on 1% agarose gels, and fragments of the expected size were excised and purified with the Monarch DNA Gel Extraction Kit (New England Biolabs, T1120L). Library concentrations were determined with the Qubit 1× dsDNA HS Assay Kit (Thermo Fisher Scientific, Q33231). Pooled libraries were sequenced by GENEWIZ (Azenta Life Sciences) to a depth of approximately 4,200×.

Quantification of protein abundance for MPRA experiments. The activity score[25] was calculated as weighted averages of bin assignments. Read counts were first normalized for library size using geometric mean-based size factors: $n_i' = \frac{n_i}{s_i}$, $s_i = \frac{N_i}{\sqrt[4]{\prod_{j=1}^{4} N_j}}$, where n_i is the raw read count, N_i is the total reads in bin i and s_i is the normalization factor. Activity score was then calculated as $\text{Activity score} = \frac{\sum_{i=1}^{4} i \times n_i'}{\sum_{i=1}^{4} n_i'}$. Sequences with activity score ≥2.0 were classified as functional IRES.

Identification and analysis of high-activity related motifs

To identify sequence features associated with high IRES activity, we performed k-mer enrichment analysis on experimentally validated sequences. Experimental sequences were derived from MPRA experiments of IRES-DM and IRES-EA libraries. Based on activity score from bicistronic luciferase reporter assays, sequences were classified into five levels: very high, high, medium, low and negative. For this analysis, we merged very high and high into a single 'high' group ($n = 518$), and used medium ($n = 8,702$) and low ($n = 14,315$) groups for activity gradient assessment. The negative group was excluded due to insufficient sample size. In addiiton, we used natural IRES sequences ($n = 7,610$) and negative control sequences (non-IRES, $n = 37,306$) from the training set as references. All sequences were restricted to 160–200 nt in length.

The identification of high-activity related motifs employed a two-step strategy. First, we identified motifs that were enriched in experimental high-activity sequences and showed clear activity dependence. For each k-mer, we calculated sequence-level frequency across groups, defined as the percentage of sequences containing the k-mer (counting once per sequence regardless of multiple occurrences). The filtering criteria included: (1) the k-mer frequency in the high group must exceed both low and negative control groups (high > low and high > negative), ensuring positive correlation with activity; (2) Fisher's exact test (one-sided) comparing high versus negative control distributions, requiring high/negative enrichment ≥1.5 and P value ≤0.05; (3) high/low gradient (ratio of high group frequency to low group frequency) ≥1.5, quantifying the motif's activity dependence. Motifs satisfying all three criteria were defined as high-activity related motifs.

Building on this foundation, we evaluated the distribution patterns of these high-activity related motifs in natural IRES to understand their biological context. We calculated the high/natural frequency ratio and classified motifs according to the following criteria: (1) if a motif had ≤5% frequency in natural IRES and high/natural frequency ratio ≥2, it was classified as design-enriched motifs, indicating specific enrichment in the proposed model-generated high-activity sequences; (2) otherwise, it was classified as natural-prevalent motifs, indicating the motif is also prevalent in natural IRES and reflects common IRES functional features learned by the model.

To assess the combinatorial effects of multiple motifs, we counted the total number of high-activity related motifs present in each experimental sequence (counting each motif once per sequence) and grouped sequences by motif count (0, 1, 2, 3, 4, 5 and 5+). Mean activity score and standard deviations were calculated for each group to evaluate the quantitative relationship between motif abundance and IRES activity.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets used in this study, including training data and evaluation data for IRES-LM, IRES-EA and IRES-DM models, along with all experimental results, are available via GitHub at https://github.com/a96123155/IRES_Prediction_Design. The archived version of the repository is available via Zenodo at https://doi.org/10.5281/zenodo.15081323 (ref. 34). Source data are provided with this paper.

Code availability

The code is available via GitHub at https://github.com/a96123155/IRES_Prediction_Design with GNU General Public License Version 3. This repository contains the code dependencies, operating environment, instructions and source code for the proposed IRES-LM, IRES-EA and IRES-DM models. The DOI is available via Zenodo at https://doi.org/10.5281/zenodo.15081323 (ref. 34).

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Author contributions

Y.C. developed the model. D. Yin, G.X., Junze Zhang, N.Z. and Y.Z. performed experimental validation. X.W. performed MPRA analysis. Y.S. and Y.L. processed data and implemented benchmark methods. D. Yu, D. Yin and G.X. initiated the experimental part of the project. H.G. and Jason Zhang reviewed the paper. M.W. and L.C. led the entire project. All authors contributed to paper preparation.

Competing interests

The authors declare no competing interests.

Additional information

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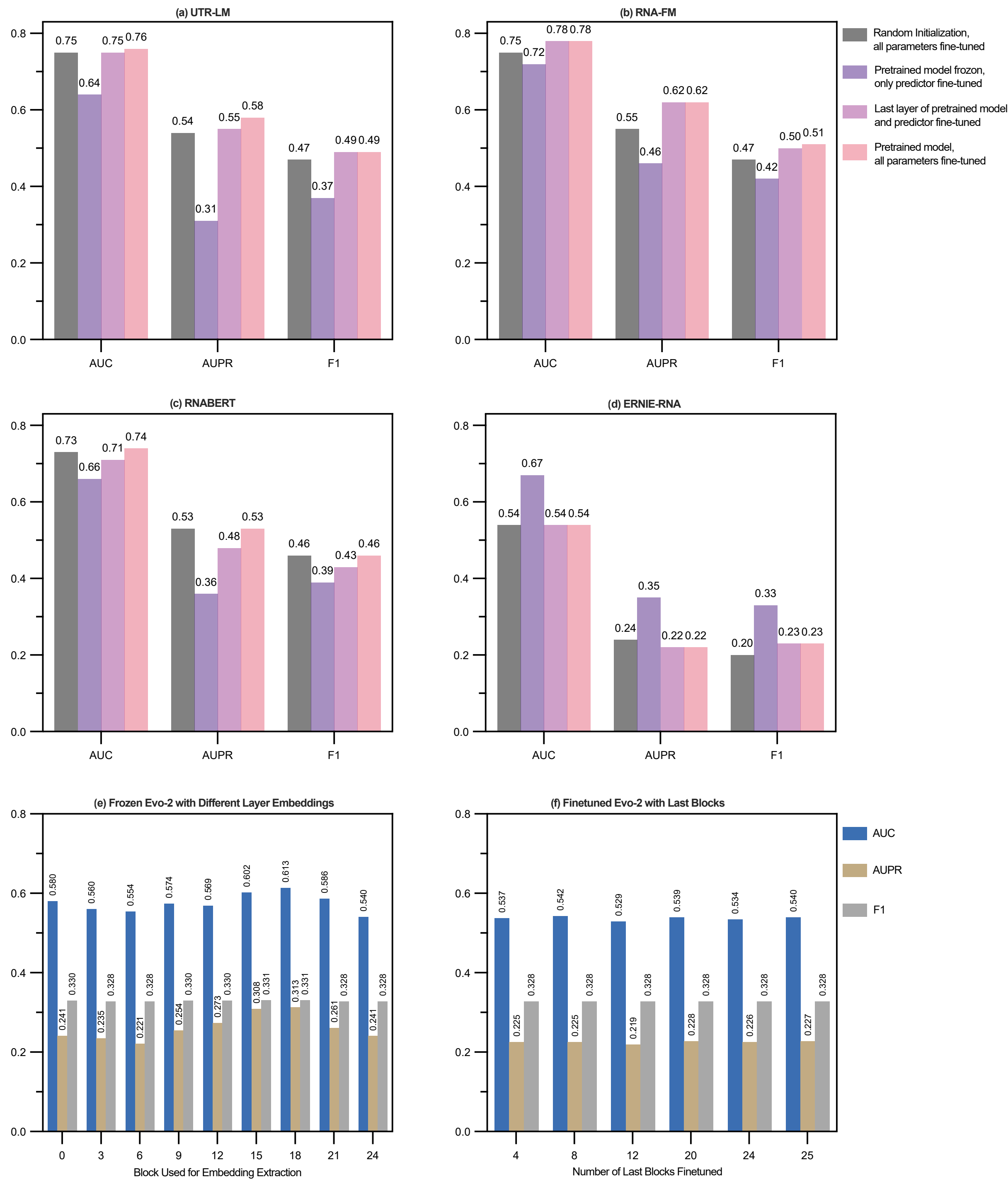

Extended Data Fig. 1 | Performance comparison of four pre-trained models across different fine-tuning strategies for IRES identification. Performance evaluation of (**a**) UTR-LM, (**b**) RNA-FM, (**c**) RNA-BERT, and (**d**) ERNIE-RNA using four different training strategies: random initialization with all parameters fine-tuned, pretrained model frozen with only predictor fine-tuned, last layer of pretrained model and predictor fine-tuned, and pretrained model with all parameters fine-tuned. (**e**) Performance of frozen Evo-2 with downstream head trained on embeddings extracted from different layer blocks. (**f**) Performance of Evo-2 with last N blocks and downstream head fine-tuned together.

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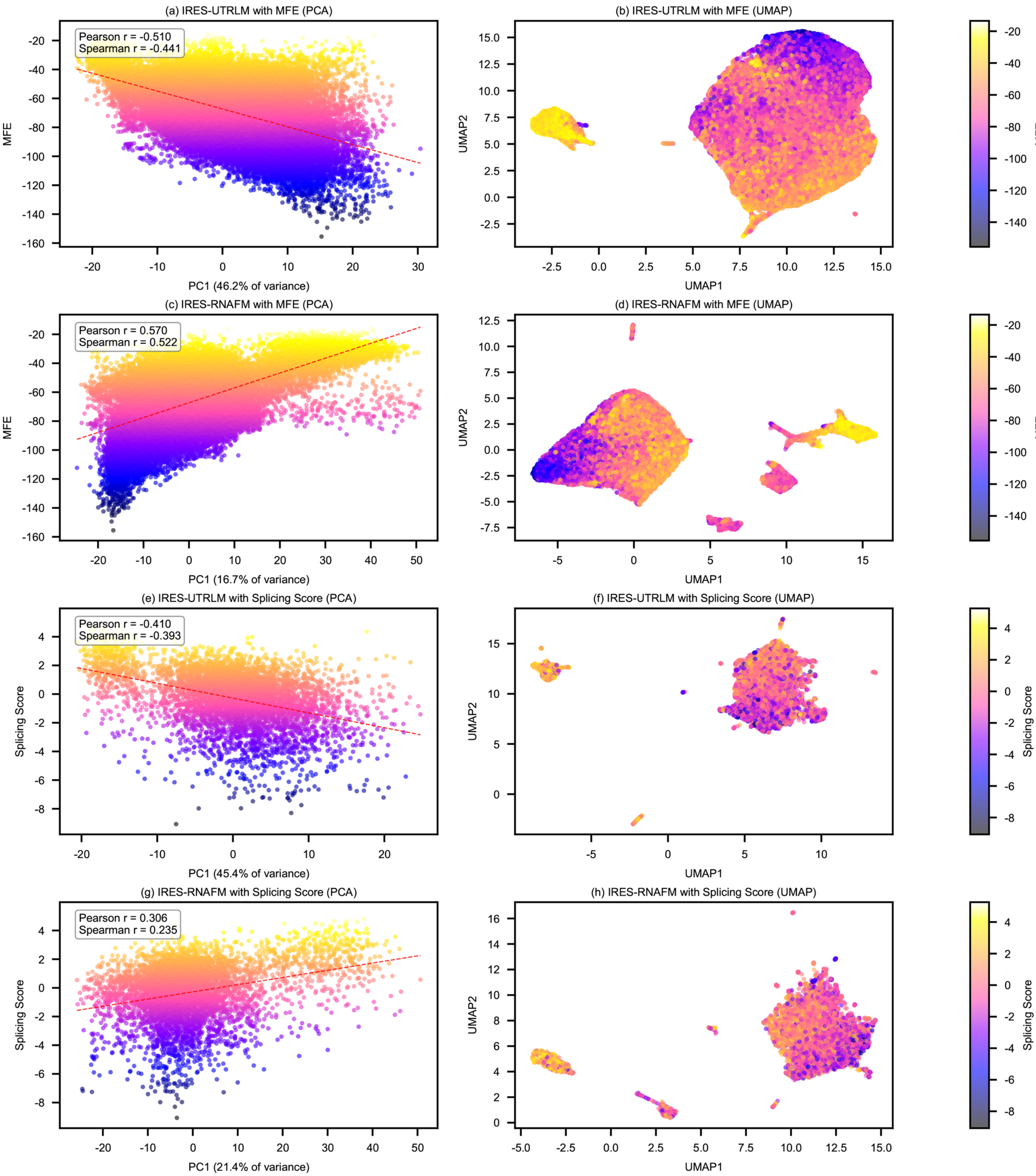

Extended Data Fig. 2 | IRES-LM representations capture biologically relevant features without explicit supervision. Analysis based on 55,000 sequences annotated with MFE values and 5,388 sequences annotated with splicing scores, with these biological properties never used during model training. **a**,**b**, IRES-UTRLM analysis with minimum free energy (MFE). **a**, Principal component analysis (PCA) showing correlation between first principal component and MFE values. The red dashed line indicates linear regression fit. **b**, Uniform manifold approximation and projection (UMAP) visualization of complete model representations colored by MFE. **c**,**d**, IRES-RNAFM analysis with (**c**) PCA and (**d**) UMAP for MFE. **e**,**f**, IRES-UTRLM analysis with (**e**) PCA and (**f**) UMAP for splicing scores. **g**,**h**, IRES-RNAFM analysis with (**g**) PCA and (**h**) UMAP for splicing scores.

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Software and code

Policy information about availability of computer code

Data collection	The study collected a dataset of binary-labeled sequences indicative of their functionality as Internal Ribosome Entry Sites (IRESs), retrieved from multiple data sources, including the study of Shira Weingarten-Gabbay et al., IRESbase, IRESite, Rfam, and the training set of IRESpred. The dataset includes 9,072 IRES and 37,602 non-IRES sequences stored in https://github.com/a96123155/IRES_Prediction_Design/blob/main/Data/v2_dataset_with_unified_stratified_shuffle_train_test_split.csv.zip, as well as 21 experimentally validated circRNA IRESs from DeepCIP in https://github.com/a96123155/IRES_Prediction_Design/blob/main/Data/circires_case.fa
Data analysis	Data analysis was performed using custom Python (v3.9/3.12) scripts implementing three frameworks: IRES-LM, IRES-EA, and IRES-DM. Model training was conducted using PyTorch on NVIDIA Tesla GPUs. Sequence clustering was performed using CD-HIT-EST (v4.8.1). All software dependencies and version numbers are specified in the repository requirements.txt file. The code is publicly available at https://github.com/a96123155/IRES_Prediction_Design under the GNU General Public License v3. The archived version is available at Zenodo (https://doi.org/10.5281/zenodo.15081323).

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"All data used in this study are publicly available in the GitHub repository (https://github.com/a96123155/IRES_Prediction_Design/). The repository contains the complete dataset used for training and testing the models, including 9,072 IRES and 37,602 non-IRES sequences in the file './Data/v2_dataset_with_unified_stratified_shuffle_train_test_split.csv.zip', as well as 21 experimentally validated circRNA IRESs collected from DeepCIP in './Data/circires_case.fa'. There are no restrictions on data availability, and no clinical or third-party datasets were used without proper citation.

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Population characteristics	N/A
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Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The study utilized a comprehensive dataset comprising 46,774 RNA sequences, including 9,072 IRES and 37,602 non-IRES sequences. This sample size was determined based on the availability of annotated IRES sequences in public databases. For individual experimental validation, we tested 37,293 non-IRES sequences for mutation potential, with subsequent experimental validation performed on selected sequences (7 EMCV IRES variants and 9 CVB3 IRES variants) to verify computational predictions. For MPRA experiments, we tested 2 libraries with 12,000 sequences, respectively.
Data exclusions	No data were excluded from the primary analyses. For the diffusion model-generated sequences, post-generation filtering was applied using pre-established criteria: absence of consecutive repeat nucleotides, lengths ranging from 160 to 200 nucleotides, and IRES predicted probability exceeding 50%. These criteria were established to ensure biological relevance of the generated sequences.
Replication	Computational predictions were validated through 10-fold cross-validation to ensure reproducibility. Individual experimental validation of selected IRES variants was performed with 2 biological replicates to confirm the computational model's predictions. All key findings can be successfully replicated followed by the GitHub repository.
Randomization	For computational analysis, the dataset was stratified and randomly split into training and testing sets to ensure balanced representation of IRES and non-IRES sequences. For experimental validation, selection of sequences for testing was based on computational predictions rather than random allocation, as the study aimed to validate specific computational hypotheses.
Blinding	Blinding was not relevant to the computational aspects of this study as the analysis was performed using automated algorithms with objective metrics. For experimental validation of IRES activity, standard laboratory protocols were followed with appropriate controls to minimize bias.

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☒	☐ ChIP-seq
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Eukaryotic cell lines

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Cell line source(s)	HEK293T cells were from ATCC.
Authentication	Cell lines were authenticated by STR profiling.
Mycoplasma contamination	Cell lines used in this work were tested negative for mycoplasma comtamination.
Commonly misidentified lines (See ICLAC register)	None.

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Seed stocks	N/A
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Methodology

Sample preparation	Cells were trypsinized and resuspended in FACS buffer (DPBS + 2% FBS).
Instrument	CytoFLEX analyzer (Beckman, USA)
Software	CytoExpert for FACS data collection; Flowjo for data analysis.

Cell population abundance	At least 10000 cells were analysis in each experiment
Gating strategy	Main cell population was first defined with FSC-A and SSC-A. Singlets werer then gated from FSC-A and FSC-H. fluorescece postive gates were defined based on untransfected cells.

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